

EXPERIMENT 3 - Modified

Fixed Throttle Performance Test of a Honda Gasoline Engine

1. Introduction

The purpose of this experiment is to measure the torque, power, specific fuel consumption (SFC), and thermal efficiency (η) of a typical automotive engine at a fixed throttle setting.

1. Torque: load on the engine [$\text{N}\cdot\text{m}$ or $\text{lbf}\cdot\text{ft}$]
2. Power: rate of work done or energy transferred [W or hp]
 - Power input determined by fuel consumption
 - Power output = torque x engine speed
3. SFC: rate of fuel consumption divided by power output of the engine
4. Thermal efficiency: power output / power input

Gasoline engines are spark-ignition internal combustion engines, which use gasoline (petrol) fuel. Gasoline engines differ from diesel engines in that they use spark plugs to ignite the air-fuel mixture in the cylinder. Diesel engines, on the other hand, utilize the heat of compression (compressed air) to ignite the injected fuel [1]. The thermodynamic cycle of a gasoline engine is referred to as the Otto cycle, named after the German engineer Nikolaus August Otto [2]:

- 0-1: Intake of the air-fuel mixture
- 1-2: Compression of the air-fuel mixture
- 2-3: Combustion after ignition by a spark plug
- 3-4: Expansion due to combustion
- 4-1: Exhaust after opening of exit valve
- 1-0: Push-out of the combustion products

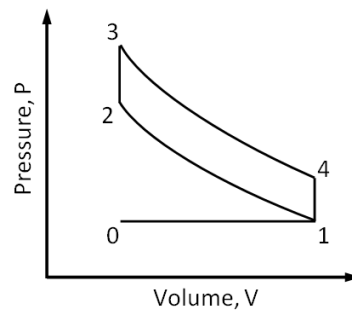


Figure 1: P-V diagram of an Air Standard Otto Cycle

The premixing of air and fuel allows gasoline engines to operate at higher speeds (higher rpm) than diesel engines. As a result, gasoline engines typically outperform diesel engines over a wider range of operating conditions. Diesel engines, on the other hand, can operate at much higher compression ratios (gasoline engines are susceptible to premature ignition at higher pressure ratios), and so they may be more efficient than gasoline engines for higher power outputs. Internal combustion engines typically have thermal efficiencies under 35%, but there is significant variance depending on fuel type, engine design, and operational parameters. Compression ratios of gasoline engines range from 9:1 to 12:1. Compression ratios of diesel engines range from 15:1 to 22:1. Students can refer to experiments 2 and 3 for more information regarding diesel engines.

The engine drives a dynamometer which is bolted to the engine block. This dynamometer is a water brake with a swinging stator; the restraining force and hence the torque, is provided by a hydraulic load cell and the load cell pressure is read on a remotely-mounted gauge. The load is varied by changing the water flow rate through the dynamometer. The engine speed is measured by a remotely mounted tachometer driven by a flexible cable. Fuel flow rate is measured by a rotameter-type flowmeter, which consists of a ball in a tapered-bore tube. Instruments showing engine coolant temperature, oil temperature and oil pressure are also mounted on the console, so that the engine operating conditions can be monitored. If any of these instruments exceed their ranges (identified by red pointers), immediately bring the engine to idle and ask for assistance from the T.A. or Lab Technician.

2. Pre-Lab

The following tasks must be performed before you come to the lab:

- 1- Prepare the front matter of your logbook (experiment name, date, reference to procedure, etc.)
- 2- Starting from Figure 2, draw a simple control volume schematic for the engine. Indicate any other fluid streams flowing through the engine, and show any energy transfers by heat or work.
- 3- Prepare Table 1 (below) in your logbook. A printout glued into your book is acceptable.

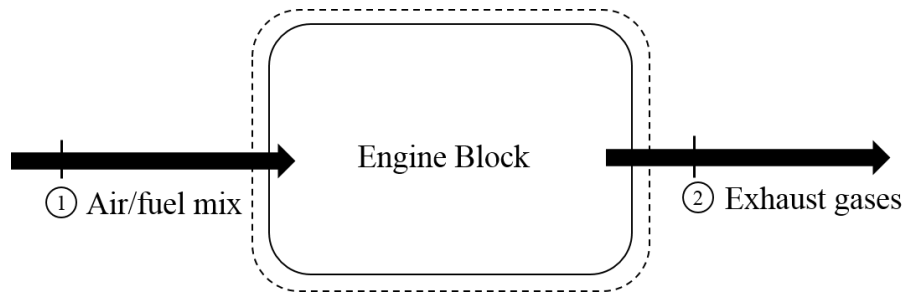


Figure 2: Incomplete control volume schematic of the engine block

3. Instructions

Table 1: Experimental Data

Test No.	Load Scale Reading	Engine Speed (rpm)	Fuel Flow Reading (s)	Oil Pressure (psi)	Oil Temp (°F)	Water Temp (°F)
1 (idle)	0.5	1700				
2	2	4000				
3		3500				
4		3000				
5		2500				
6		2000				
7 (idle)	0.5	1700				

***** Always increase RPM (engine speed) by opening the throttle prior to increasing load (torque) for step 3 below. *****

***** Use the OUTER scale on torque gauge. *****

- 1) Have an instructor initial the data sheet.
- 2) Allow the engine to warm up under a light load (1700 rpm and 0.5 on the OUTER scale of the torque gauge) before proceeding with the test. Record a set of data at this condition: engine idling point.
- 3) Set the throttle at about 3/8 throttle. To do this, open the throttle until the engine speed is 4000 rpm and increase the load to 2 on the torque gauge (OUTER scale). You may need to readjust the throttle after increasing the load to maintain the desired engine speed of 4000 rpm.
- 4) From this step on, do NOT adjust the throttle until step (8).
- 5) Record a set of data: engine speed, load scale reading, fuel flow reading, oil pressure, oil temperature and water temperature. Call the laboratory demonstrator if these instruments indicate an engine malfunction.

- 6) Increase the load until the engine speed falls to 3500 rpm and record a second set of data.
- 7) Progressively increase the load to reach the following engine speeds: 3000, 2500 and 2000 rpm, recording data at each speed.
- 8) Simultaneously reduce the load to 0.5 and close the throttle to reduce the engine speed to 1700 rpm. Record one last set of data. Check your data to make sure that there are no outliers.
- 9) Do NOT shut down the engine. Have the instructor check and initial the data sheet. Clean up and leave the equipment in good order.

4. Calculations & Presentation of Results

- 1) Calculate (and tabulate in a Results table) torque ($T, [lb_f \cdot ft]$), power output of the engine ($P_{out}, [hp]$), fuel mass flow rate ($\dot{m}_f, [\frac{lb_m}{hr}]$) and specific fuel consumption ($SFC, [\frac{lb_m}{hp \cdot hr}]$) for all data points. The specific gravity (S.G.) of gasoline fuel is 0.68.
- 2) For the point of best SFC, estimate the power produced by the combustion of air and fuel in the engine ($P_{in}, [hp]$). The lower heating value (LHV) of the gasoline fuel may be taken as 19,000 Btu/lbm (44 MJ/kg). The LHV is the amount of heat (energy) that can be generated through complete combustion of a unit mass of fuel (combustion products at 150°C), assuming all water in the exhaust stream is in vapour form.
- 3) Calculate the thermal efficiency (η_{th}) of the engine at the point of best SFC.
- 4) On a graph with engine speed on the horizontal axis, plot the calculated torque, output power and SFC; use different symbols for each quantity and pass reasonably smooth curves through the data. Title the graph and label the axes and the curves. (Do not use the same vertical axis for all three curves, see the graph on page 4 of this lab manual as an example. Note that SFC uses the axis on the right-hand side). Do not plot the idle points.

5. Discussion Questions

- 1) What does “fixed throttle” mean with respect to the engine’s operation? Hint: This Honda is fuel injected, so the fuel flow is electronically controlled. Think of what else the engine needs that the throttle could be controlling.
- 2) Discuss the thermal efficiency of the engine. Is it reasonable? Referring to your schematic, suggest areas where energy may have been lost.
- 3) Is it desirable to have a high or low SFC? How do SFC and fuel consumption ($\dot{m}_f$) differ?
- 4) Discuss trends in your performance graphs (torque, power and SFC), referring to the relationships between torque, speed and power.
- 5) Briefly discuss the suitability of this type of engine in a motor vehicle considering your results. Is the performance similar at all engine speeds? Remember that a motor vehicle is expected to function through a range of speeds (which can be extended using a gearbox).

6. References

- [1] Moran, M.J. & Shapiro, H.N. (2007) “Fundamentals of Engineering Thermodynamics”, 6th Edition, Wiley.
- [2] Müller, I. & Müller, W.H. (2009) “Fundamentals of Thermodynamics and Applications with Historical Annotations”, Springer.

7. Other Sources

- Obert, "Internal Combustion Engines" (TJ 785-02)
- Schmidt, "The Internal Combustion Engine" (TJ 785S3)
- Ricardo and Hempson, "The High Speed Internal Combustion Engine" (TJ 785R52)

Important Equations and Constants:

$$T [lb_f \cdot ft] = 5.252 \times scale\ reading$$

$$SFC \left[\frac{lb_m}{hp \cdot hr} \right] = \frac{\dot{m}_f \left[\frac{lb_m}{hr} \right]}{P_{out} [hp]}$$

$$P_{out} [hp] = scale\ reading \times \frac{engine\ speed [rpm]}{1000}$$

$$\eta_{th} [\%] = \frac{P_{out}}{P_{in}}$$

$$P_{in} = LHV \times \dot{m}_f$$

$$Specific\ Gravity: \quad S.G. = \frac{\rho_{fuel}}{\rho_{H_2O}}$$

$$\rho_{H_2O} = 1.0\ g/ml$$

$$1\ kg = 2.2\ lb_m$$

$$1\ hp = 2544\ BTU/hr$$